Unit 6

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① What are the main goals of implementing IOT in manufacturing environment ?

⇒ ① IOT aims to connect machines, sensors & systems. It enables real-time monitoring and automation. Through connected technologies and intelligent systems can achieve better control.

② The main goal is :-

a) Data-driven Optimization :-

- In IOT environment, sensors are attached to collect real-time data such as temperature, vibration, pressure.
- This data is then analyzed using analytics tools to predict failures before they actually occur
- This enables scheduled or predictive maintanence instead of costly, unplanned repairs. & extend machine life

ex CNC machine equipped with vibration & temp predicts wear and alerts maintainance before breakdown.

b) Automation & Efficiency :-

- IOT removes repetitive operations by automating key manufacturing process
- It streamlines workflows and minimize errors by removing human intervention.
- This improves overall produc^{n} efficiency & speed.

example → In car, robots automatically weld parts while sensors cordinate movements to prevent overlap or idle time.

c) Supply Chain Management :-

- IOT enables tracking of raw materials, components & finished goods throughout production cycle.
- Automatic alerts and reorder systems reduce stockouts and delays.
- This improves logistics efficiency, inventory management & customer satisfication

ex. A smart warehouse uses IOT sensors to monitor pallet movement & automatically reorder materials

d) Enhanced safety :-

- IOT-enabled devices including wearable sensors to detect unsafe working condⁿ.
- These track workers movements, temperature and gas levels in hazardous zones.
- Alerts notify if unsafe parameters are detected and ensure safer work environment

ex. workers in chemical plant wears smart helmets that detect toxic gas leaks & triggers.

e) Foundation for Industry 4.0 :-

- IIOT is core component of Industry 4.0
- It focuses on integration of physical & digital system in industrial process
- This connected system creates "smart factories" capable of automatic decision making.

ex. Smart factory machines adjust prodⁿ speed based on live demand data & ERP systems

(7)

(2) How does IOT help improve overall equipment effectiveness (OEE)?

⇒ (1) OEE is key performance metric used in manufacturing to measure how effectively equipment is being utilized.

(2) IOT helps improve OEE by enabling real-time monitoring & data-driven optimization & praesses

(3) Through connected sensors & analytics, IOT provides continuous insights into availability, performance & quality

OEE = Availability x Performance x Quality.

1) Availability → Reducing Unplanned Downtime

- IOT sensors continuously track equipment cond^n such as vibrations, temp & pressure
- Using this algo can detect early signs of failures
- System alerts operators before breakdown occurs allowing maintence & preventing sudden stopage.
- It ensures machines remain operational for longer periods.

Result → less unplanned downtime & reduce repair costs

2) Performance → Optimizing Throughput

- IOT devices collect live operational data on speed, energy consumption & machine cycles
- Analytics tools process this data to identify inefficiencies
- Operators can make real-time adjustments to improve machine throughput.

- As a result overall prod° efficiency & speed are improved.

Result:- Higher performance & better utilization of capacity

(3) Quality:- Reducing Defects & Rework

- IOT enables continuous monitoring for prod° process that effect product quality
- Smart sensors & cameras can detect defects instantly during manufacturing
- Data analytics tools help identify process variations that lead to quality issues.
- Reduces rework, material waste & customer complaints

Result:- Improved quality & higher customer satisf°

(4) Data-Driven Decision Making

(5) Integ° with Enterprise System

2 diagrams from PPT

(Q.3) Outline the structure of Industrial Automation & control system (IACS) reference model?

→ ① IACS Reference Model defines a layered structured that organized devices, systems and networks used in Industrial automation.
② Each level has specific roles - from physical control at lowest level to enterprise management at the top.
③ It also supports integration betn operational technology (OT) & IT in efficient way.

* Level 0 → Process or equipment under control
- It consists of sensors & actuators involved in manufacturing process.
- These IOT devices perform IACS funcn such as moving a manufacturing robot, spraying & welding.
- These devices are in commn with basic control devices at level 1.

* Level 1 :- Basic control
- It includes controllers/PLC's that provide basic control
- Execute logic & safety funcn based on sensor i/p from level 0
- This provides basic automation & protection for industrial equipment.
- Serves as direct interface betn machine & supervisory system.

* Level 2 :- Supervisory control
- Includes supervisory control funcn and devices such as HMIs.

- Engg. can view adjust & control plant operations in real time.
- Also includes histoians, applicn severs & engg database etc.

*Level 3 :- Plant control & operations
- Includes SCADA systems, file severs & prodn scheduling software
- Helps ensure plant runs efficiently acc to prodn goals.

*Level 4 :- Enterprise system.
- Represent bussiness & enterprise layers of IACS model
- Include system like ERP, email & file services,
- Ensures bussiness objectives & customers demands.

• Demilitarized zone (DMZ):-
→ Is the CPwE betn plant network & toadn network.
→ DMZ security is critical to plant operations as it protects machines at lower level

• Enterprize zone :-
→ Level 4 & 5 in enterprise zone relate to toadn IT / enterprise networking funcn like internet, email, file services etc.

Q.5 Compare EtherNet/IP, PROFINET & Modbus/TCP in terms of architecture, speed and interrupt probability.

⇒

a) EtherNet/IP :-

- Ethernet Industrial Protocol is an implementation of Common Industrial Protocol (CIP) over Ethernet and TCP/IP.
- It combines robustness of CIP protocol with adoption of Ethernet networking.
- Allows real-time control & data exchange over standard Ethernet.
- Highly scalable supporting both small and large automation system.
- Uses multicast comm^{n} to share info with multiple devices.

b) PROFINET (Process Field Net) :-

- It is an industrial Ethernet standard designed for fast, deterministic comm^{n} in automation
- Exchanges data directly at applic^{n} layer avoiding TCP/IP delays
- Includes PLC's, fields devices & HMI's for visualization & control
- Offers Media Redundancy Protocol (MRP) for continuous comm^{n} even during link failures.

c) Modbus/TCP :-

- It is an Ethernet-based adaptation of traditional Modbus serial protocol
- Operates in master/slave config, where master requests data from slave devices
- Protocol is simple, open & widely implemented in various industries

- Uses TCP/IP for comm^{n} which makes it compatible with many n/w devices.

Comparison

Parameter	EtherNet/IP	PROFINET	Modbus/TCP
① Architecture	Based on CIP over standard Ethernet & TCP/IP	Direct applic^{n} layer Ethernet betⁿ controllers & devices	Master/slave model using TCP/IP of Modbus
② Speed	High-speed real time capable scalable	Very high speed deterministics low latency	Moderate spee affected by TCP IP delay
③ Interoperability	Excellent-open standard, seamless IT/OT integration	High but mainly among PI supported devices	Good-open & simp but limited advanced feature

Conclusion

a) Profinet Provides fastest comm^{n} while EtherNet/IP offers excellent scalability & interoperability

b) Modbus/TCP is simpler but slower & less efficient

Q6 What is REP and how does it improve its network reliability.

→ • Intro:-

(1) REP is Cisco proprietary redundancy protocol developed for industrial Ethernet ring topologies.

(2) Primarily used in Cell/Area zone of industrial networks

(3) It ensures high speed recovery during link or switch failures

(4) REP forms logical segment in n/w, identified by unique seg ID and one port within ring is configured as an alternate (blocking) port to prevent loops.

(5) It is a layer 2 protocol that eliminates need of Spanning Tree Protocol (STP).

• Purpose :-

Main goal is to increase network availability by offering fast recovery (<50 ms) in ring topologies within the Cell/Area Zone.

• Key features :-

1) Loop prevention → Via alternate port similar to STP but faster & more scalable.

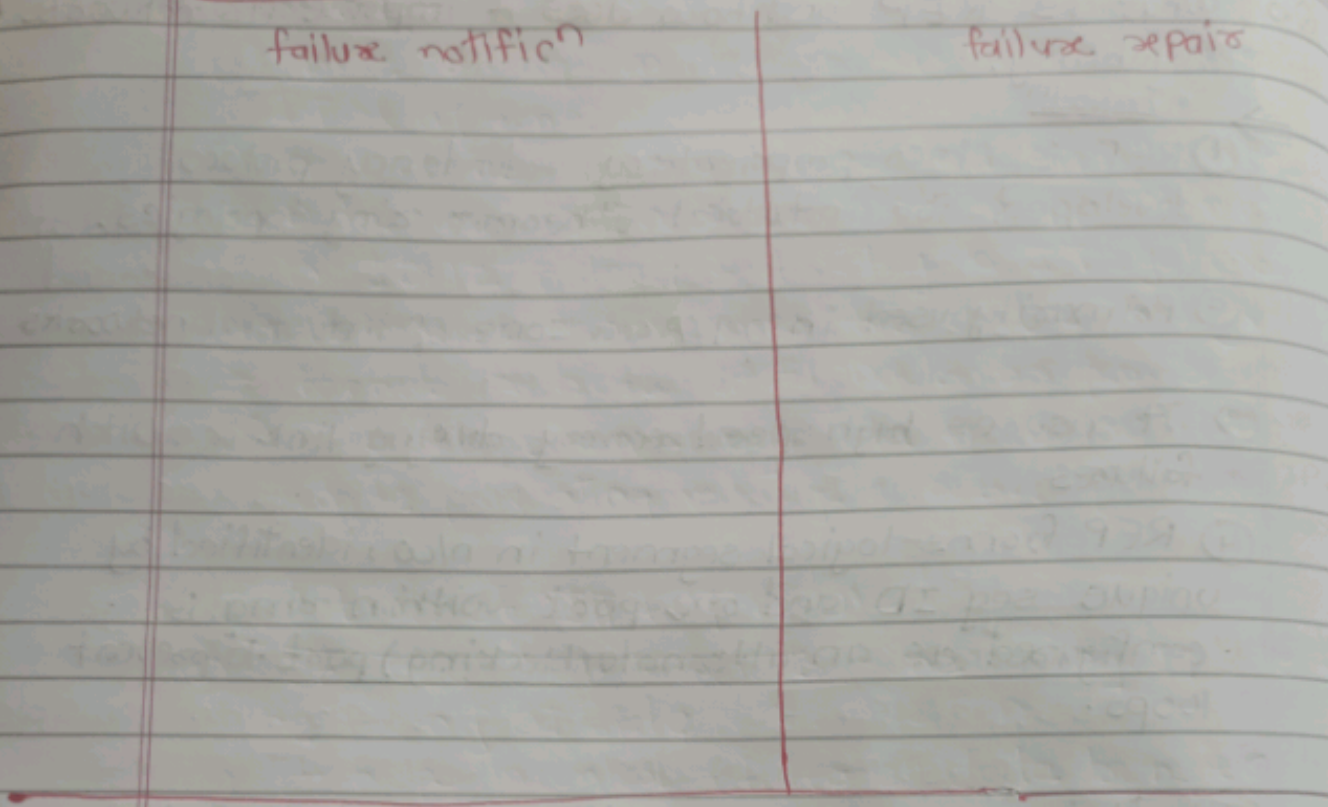
2) Supports unlimited nodes (vs. STP's 7-device limit)

3) Uses control VLAN for REP messaging.

4) Master Node manages segment behavior.

5) loss of signal (LOS) detection triggers fast convergences.

Diagram



- Working :-
  1) RFP forms ring topology in which 1 port is kept as an alternate port
  2) When link/switch failure occurs RFP quickly detects loss of signal & unblocks alternate port to restore
  3) Mast Node coordinates using control msg exchanges through VLAN
  4) Once fault is repaired RFP automatically re-blocks alternate port.

- How RFP improves Nlw Reliability.
  1) Provides fast & automatic recovery from nlw faults, minimizing downtime during failure
  2) Maintains loop-free redundancy ensuring reliable data flow
  3) Loss of signal(LOS) detection guarantee uninterrupted comm^n.

4) Overall REP improves n/w reliability & performance by reducing costs, maintaining throughput & ensuring sustained operations.

Q7 What is an IDMZ and how does it isolate IT & OT networks.

→ ① IDMZ is a secure n/w layer that acts as a buffer or intermediary betn IT.
② It is often referred to as level 3.5 in Industrial Automation & Control system (IACS) reference model
③ It is designed to protect critical industrial assests from cybersecurity threats.

⊙ Purpose :-

- Main purpose is to enforce data security policies betn enterprise zone (IT) & Industrial zone (OT)
- Allows controlled data sharing for reporting & remote monitoring.

⊙ Key features

1) Segregated N/w layers - Positioned betn enterprise & industrial n/w to act as neutral zone
2) Security infrastructure - Include firewalls, ~~reverse~~ VPN servers
3) Scalability and Compatibility → Supports integration of industrial applicn & s/w securely.

Working :-

1. IDMZ is implemented as seperate n/w layer between IT & OT zones
2. It contains security devices and applicn servers
3. Data from OT n/w is sent to applicn mirrors in IDMZ instead of directly to IT systems.
4. Enterprise n/w accesses this data through channels such as VPN tunnels or proxies
5. This allows one-way data flow for monitoring while preventing insecure access.

- Isolation of IT & OT n/w :-

1. It use dual firewalls - one facing IT side & one facing OT side
2. It helps to control & filter all data transfer
3. Only specific & approved traffic is allowed through IDMZ
4. Devices in IDMZ ensure IT users never connect directly to live industrial system
5. This structure protects OT assets from attacks ensuring both security & operational continuity.

Applic^n (online).

(8) Define Edge computing & explain its relevance in IIOT environment?

→ (1) Edge computing is a distributed computing model where data is processed close to it's source - such as sensors, controllers - instead of sending all info to centre.
(2) Traditionally factories used PC's on plant floors to collect machine data but created challenges.
(3) This setup made it difficult to effectively use & analyze large amount of prodn data.

Key components :-

1) Edge Appliance :-
- Rugged, industrial-grade comp → collects & process data
- Open source O.S like Linux

2) Hardware Adaptors :-
- Installed on machines to define I/O tags & COS data

3) Commn Protocols :- for connectivity betn machines
XML or HTTP

4) Connectivity layer :- Provides secure commn betn edge devices or cloud platforms. like SCADA or cloud platform

~~Key features~~ :-

Working :-
(1) Machines on plant floor generate massive amount of data
(2) Hardware adaptors detects these changes & send them to Edge appliance for processing
(3) Edge device analyzes & filters data locally using OS
(4) Only relevant info is transmitted to data center for long-term analysis.

Diagram

(5) This results in low latency & reliable insights even during n/w disruptions.

Relevances in IIOT environment:-

(1) Real-time Data Processing :-

⇒ It analyzes data locally at machine level enabling instant response in IIOT systems.
⇒ faster decision-making in critical operations

(2) Reduced Latency & Network load :-

Since only filtered data is sent to cloud it reduces bandwidth usage & n/w congestion.

(3) Enhanced security & Reliability

By keeping data within plant n/w edge computing minimizes cybersecurity risks & ensure continuous operations without fails.

(4) Interoperability with IP :-

Support multiple industrial standards as MTConnect OPC UA & PackML.

(5) Support for Industry 4.0 :-

forms foundation of smart factories, connecting physical & digital systems for intelligent manufacturing